

Stock Screener — Beaten-Down Value with Upside

Institutional-Grade Equity Research Tool for Claude Code

Overview

This project instructs Claude Code to act as an institutional equity research analyst: screen for stocks that have sold off significantly but retain strong fundamentals, score them on a weighted rubric, and output a professional hedge-fund-quality Excel model.

Strategy: Contrarian value with defined catalysts

Macro view: Bearish US credit and economy — prefer recession-resistant industries

Output: `output/stock_screener.xlsx` — 8-sheet institutional model

Quick Start

```
# Claude Code will do this automatically, but for reference:

# Step 1: Claude Code conducts web research and populates:
#   output/research_data.json

# Step 2: Build the Excel model
pip install openpyxl
python scripts/build_model.py

# Step 3: Recalculate formulas (MANDATORY)
python scripts/recalc.py output/stock_screener.xlsx
```

Project Structure

```
stock-screener/
├── CLAUDE.md           ← Claude Code reads this first (mission + in
├── README.md          ← This file
└──
```

```
|— examples/
|   └─ successful_trades.json      ← 3 annotated example trades with entry logi
|
|— config/
|   └─ scoring_rubric.json        ← Detailed 6-criterion scoring guide with ru
|
|— scripts/
|   └─ build_model.py             ← Excel model builder (openpyxl)
|
└─ output/
    └─ research_data_template.json ← Schema Claude Code populates with real dat
    └─ stock_screener.xlsx        ← FINAL OUTPUT (generated by build_model.py)
```

What Claude Code Must Do

Phase 1: Research

1. Read `CLAUDE.md` for full mission and strategy context
2. Read `examples/successful_trades.json` to internalize the investment pattern
3. Read `config/scoring_rubric.json` to understand scoring criteria
4. Always start with **NVO** — research current price, all catalysts, pipeline, IRA impact
5. Screen for 4–6 additional stocks using the hard filters
6. For each passing stock: gather all quantitative + qualitative data
7. Score each stock on all 6 criteria
8. Populate `output/research_data.json` (use `output/research_data_template.json` as schema)

Phase 2: Build Model

1. Run `python scripts/build_model.py`
 2. Run `python scripts/recalc.py output/stock_screener.xlsx`
 3. Verify zero formula errors
 4. Spot-check key cells for accuracy
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Excel Output — 8 Sheets

Sheet	Content
SUMMARY DASHBOARD	Master ranking table, traffic light scoring, portfolio construction guidance
STOCK SCORES	Full 6-criterion scoring matrix with weighted composite and justifications
VALUATION COMPS	Current multiples vs. peers/history; bear/base/bull price targets
FUNDAMENTAL DATA	Revenue, EBITDA, EPS, balance sheet, cash flow per stock
CATALYST TRACKER	All catalysts with timeline, probability, and price impact estimates
RISK MATRIX	Bull/base/bear scenarios with expected value calculation
NVO DEEP DIVE	Full deep dive: selloff timeline, pipeline table, IRA model, verdict
ASSUMPTIONS	All model assumptions, peer groups, data sources

Scoring Rubric (Summary)

Criterion	Weight	What It Measures
Valuation	25%	How cheap vs. history/peers/DCF
Balance Sheet	20%	Can it survive the downturn?
Catalyst	20%	Are there specific, time-bound recovery drivers?
Selloff Quality	15%	Company-specific overreaction vs. structural decline
Industry Protection	10%	Recession-resistance of core demand
Analyst Conviction	10%	Consensus rating and target momentum

Score interpretation: 7.5+ = Strong Buy | 6.5–7.4 = Buy | 5.0–6.4 = Speculative | <5.0 = Pass

Successful Trade Examples (Reference)

Ticker	Entry	Current	Return	Pattern
BA	\$190	\$210	+10.5%	Operational crisis, intact franchise, new CEO catalyst
VRTX	\$400	\$460	+15.0%	Non-core miss, monopoly core franchise, clean balance sheet
GNRC	\$140	\$204	+45.7%	Post-cycle air pocket, structural demand misread

Formatting Standards

- **Blue text:** Hardcoded inputs
 - **Black text:** Formulas
 - **Green text:** Cross-sheet links
 - **Header:** Dark navy (#1F3864) with white text
 - **Traffic lights:** Green / Yellow / Red based on composite score
 - **Zero formula errors** required before final output
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Important Notes

- Do NOT fabricate data. Mark unavailable data "N/A — not found" in yellow
- All analyst targets should be within 30–60 days of model date
- Write commentary at sell-side research report quality — this is for a sophisticated institutional investor
- NVO is the primary focus; give it the most thorough treatment
- The investor is bearish on macro — weight industry protection heavily when making final recommendations